NHANES Body Measurements Analysis

CORIZO Capstone Project-1 : Working with NumPy Matrices

1. Introduction

This report presents an exploratory and statistical analysis of body measurement data from the National Health and Nutrition Examination Survey (NHANES). The objective of this analysis is to compare anthropometric characteristics between adult male and female participants using numerical summaries, visualisations, and derived health indicators.

All computations are performed using NumPy, and visualisations are created using Matplotlib. The results are interpreted from a statistical and health-related perspective.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
import pandas as pd
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

1. Dataset Acquisition

The datasets used in this analysis are obtained from the National Health and Nutrition Examination Survey (NHANES). NHANES provides reliable and widely used health-related data collected from adult participants.

Two datasets are used:

- One containing body measurements of adult male participants
- One containing body measurements of adult female participants

The datasets were downloaded from an official public repository and stored locally for further analysis.

2. Data Loading and Conversion to NumPy Matrices

In this section, the NHANES datasets for adult males and females are loaded from CSV files into Python. The data is first read using pandas to ensure proper parsing of the files. The resulting DataFrames are then converted into NumPy matrices to enable efficient numerical computations in subsequent analyses.

```
# File paths
male_path = "/content/drive/MyDrive/Colab Notebooks/csv files/nhanes_adult_male_bmx_2020.csv"
female_path = "/content/drive/MyDrive/Colab Notebooks/csv files/nhanes_adult_female_bmx_2020.csv"

# Reading the CSV files into pandas DataFrames
male_df = pd.read_csv(male_path, sep=',', comment='#', header=0)
female_df = pd.read_csv(female_path, sep=',', comment='#', header=0)

# Convert DataFrames to NumPy matrices
male = male_df.to_numpy()
female = female_df.to_numpy()

# Displaying the shape of the matrices
male.shape, female.shape
```

```
((4081, 7), (4221, 7))
```

The datasets were successfully loaded and converted into NumPy matrices named `male` and `female`. Each row represents an individual participant, and each column corresponds to a specific body measurement. The shapes of the matrices confirm that the data has been imported correctly and is ready for further analysis.

Verification of Data Structure

To verify that the datasets have been correctly loaded and that each column corresponds to the intended body measurement, a subset of the data is displayed with descriptive column labels.

```
column_names = [
    "Weight (kg)",
    "Height (cm)",
    "Upper arm length (cm)",
    "Upper leg length (cm)",
    "Arm circumference (cm)",
    "Hip circumference (cm)",
    "Waist circumference (cm)"
]

male_display = pd.DataFrame(male, columns=column_names)
female_display = pd.DataFrame(female, columns=column_names)

male_display.head()
```

	Weight (kg)	Height (cm)	Upper arm length (cm)	Upper leg length (cm)	Arm circumference (cm)	Hip circumference (cm)	Waist circumference (cm)
0	98.8	182.3	42.0	40.1	38.2	108.2	120.4
1	74.3	184.2	41.1	41.0	30.2	94.5	86.8
2	103.7	185.3	47.0	44.0	32.0	107.8	109.6
3	86.0	167.8	39.5	38.4	29.0	106.4	108.3

```
female_display.head()
```

	Weight (kg)	Height (cm)	Upper arm length (cm)	Upper leg length (cm)	Arm circumference (cm)	Hip circumference (cm)	Waist circumference (cm)
0	97.1	160.2	34.7	40.8	35.8	126.1	117.9
1	91.1	152.7	33.5	33.0	38.5	125.5	103.1
2	73.0	161.2	37.4	38.0	31.8	106.2	92.0
3	61.7	157.4	38.0	34.7	29.0	101.0	90.5

The displayed rows confirm that each dataset consists of seven columns representing the specified body measurements. The structure of the data is consistent with the provided documentation and suitable for further analysis.

3. Distribution of Body Weight

To examine and compare the distribution of body weight among male and female participants, histograms are constructed. Both histograms are plotted using identical x-axis limits to ensure a fair and consistent visual comparison between the two distributions.

```
female_weights = female[:, 0]
male_weights = male[:, 0]

min_weight = min(female_weights.min(), male_weights.min())
max_weight = max(female_weights.max(), male_weights.max())

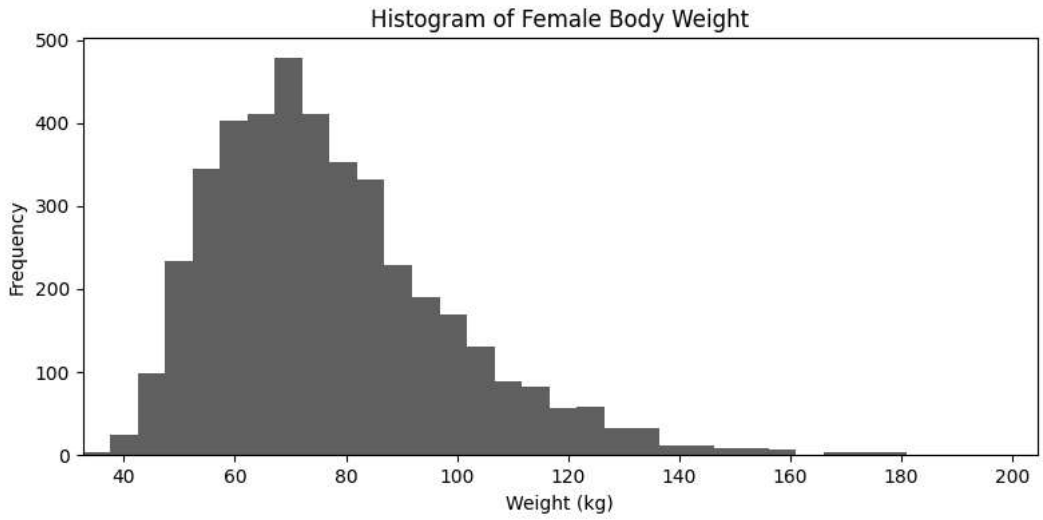
# Create a figure with two subplots (stacked vertically)
plt.figure(figsize=(8, 8))

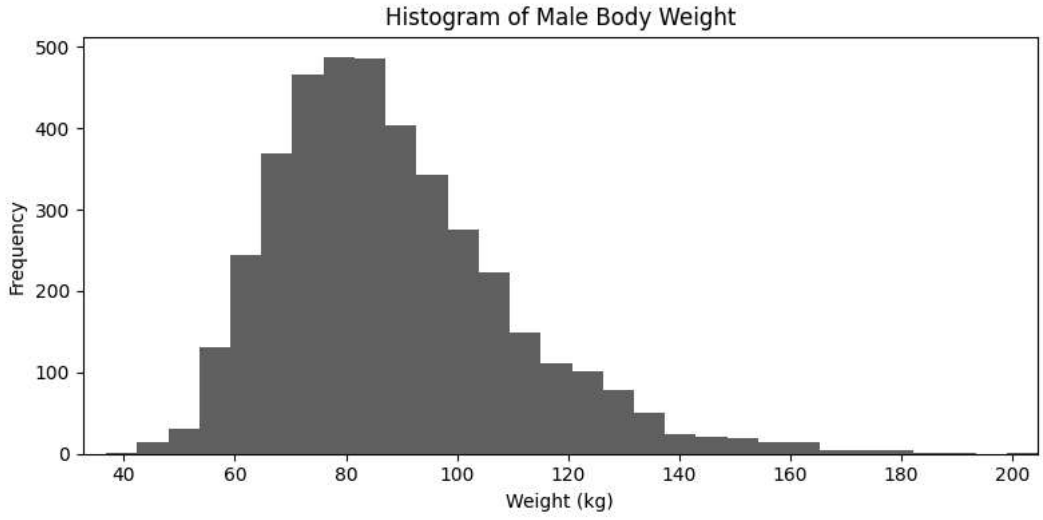
# Female weight histogram (top)
plt.subplot(2, 1, 1)
```

```
plt.hist(female_weights, bins=30)
plt.title("Histogram of Female Body Weight")
plt.xlabel("Weight (kg)")
plt.ylabel("Frequency")
plt.xlim(min_weight, max_weight)

# Male weight histogram (bottom)
plt.subplot(2, 1, 2)
plt.hist(male_weights, bins=30)
plt.title("Histogram of Male Body Weight")
plt.xlabel("Weight (kg)")
plt.ylabel("Frequency")
plt.xlim(min_weight, max_weight)

# Adjust layout
plt.tight_layout()
plt.show()
```





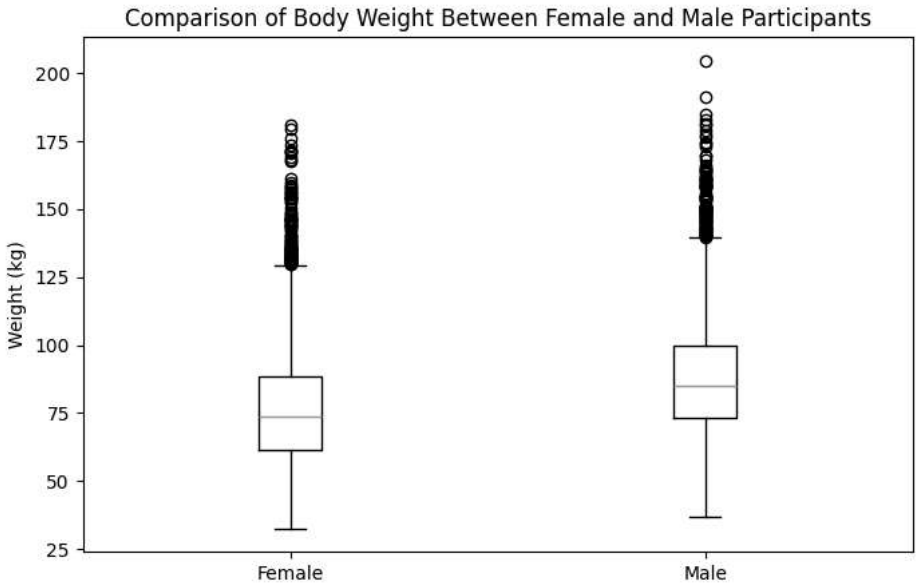
The histograms reveal clear differences in the distribution of body weight between male and female participants. Male participants generally exhibit higher body weights and a wider spread, indicating greater variability. Female weights appear more concentrated around lower values. Both distributions show slight right skewness, suggesting the presence of higher-weight outliers.

4. Boxplot Comparison of Body Weight

A box-and-whisker plot is used to compare the distributions of body weight between male and female participants. This visualization highlights differences in median values, variability, and the presence of outliers.

```
# Create a box-and-whisker plot comparing female and male weights
plt.figure(figsize=(8, 5))
plt.boxplot([female_weights, male_weights], labels=["Female", "Male"])
plt.ylabel("Weight (kg)")
plt.title("Comparison of Body Weight Between Female and Male Participants")
plt.show()
```

```
/tmp/ipython-input-1449596037.py:3: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_lab
  plt.boxplot([female_weights, male_weights], labels=["Female", "Male"])
```



The boxplot shows that male participants have a higher median body weight compared to female participants. The interquartile range for males is wider, indicating greater variability in body weight. Both distributions contain outliers, suggesting the presence of individuals with unusually high or low body weights.

5. Numerical Summary of Body Weight

To quantitatively describe and compare the body weight distributions of male and female participants, several numerical aggregates are computed. These include measures of location, dispersion, and shape, which provide a statistical foundation for the visual observations made earlier.

```
from scipy.stats import skew, kurtosis

# Female weight statistics
female_mean = np.mean(female_weights)
female_median = np.median(female_weights)
female_var = np.var(female_weights)
female_std = np.std(female_weights)
female_skew = skew(female_weights)
female_kurt = kurtosis(female_weights)

# Male weight statistics
male_mean = np.mean(male_weights)
male_median = np.median(male_weights)
male_var = np.var(male_weights)
male_std = np.std(male_weights)
male_skew = skew(male_weights)
male_kurt = kurtosis(male_weights)

stats_df = pd.DataFrame({
    "Statistic": ["Mean", "Median", "Standard Deviation", "Skewness", "Kurtosis"],
    "Female": [
        round(female_mean, 2),
        round(female_median, 2),
        round(female_std, 2),
        round(female_skew, 2),
        round(female_kurt, 2)
```

```
    ],
    "Male": [
        round(male_mean, 2),
        round(male_median, 2),
        round(male_std, 2),
        round(male_skew, 2),
        round(male_kurt, 2)
    ]
})
stats_df
```

	Statistic	Female	Male
0	Mean	77.40	88.36
1	Median	73.60	85.00
2	Standard Deviation	21.54	21.42
3	Skewness	1.03	0.98
4	Kurtosis	1.40	1.47

The numerical summaries support the visual findings from the histograms and boxplots. Male participants exhibit higher mean and median body weights, as well as a larger standard deviation, indicating greater variability. The positive skewness values confirm slight right skewness in both distributions, which is consistent with the presence of higher-weight outliers.

6. Body Mass Index (BMI)

Body Mass Index (BMI) is a commonly used indicator to assess body mass relative to height. In this step, BMI is computed for all female participants using their weight and height measurements and added as an additional column to the female dataset.

```
# Re-initialize female array from original DataFrame to prevent multiple appends
female = female_df.to_numpy()

# Extract weight and height from the female matrix
female_weight = female[:, 0]
female_height_cm = female[:, 1]

# Convert height from cm → meters
female_height_m = female_height_cm / 100

# Computing BMI
female_bmi = female_weight / (female_height_m ** 2)

#Appending BMI as the 8th column
female = np.column_stack((female, female_bmi))

# Verification of the result
female.shape

# Displaying the updated Data Frame
column_names_bmi = [
    "Weight (kg)",
    "Height (cm)",
    "Upper arm length (cm)",
    "Upper leg length (cm)",
    "Arm circumference (cm)",
    "Hip circumference (cm)",
    "Waist circumference (cm)",
    "BMI"
]

female_display = pd.DataFrame(female, columns=column_names_bmi)
female_display.head()
```

	Weight (kg)	Height (cm)	Upper arm length (cm)	Upper leg length (cm)	Arm circumference (cm)	Hip circumference (cm)	Waist circumference (cm)	BMI
0	97.1	160.2	34.7	40.8	35.8	126.1	117.9	37.835041
1	91.1	152.7	33.5	33.0	38.5	125.5	103.1	39.069720
2	73.0	161.2	37.4	38.0	31.8	106.2	92.0	28.092655
3	61.7	157.4	38.0	34.7	29.0	101.0	90.5	24.904378

7. Standardisation of Female Body Measurements

To ensure comparability across variables with different units and scales, the female dataset is standardised by computing the z-scores of each column. The resulting matrix, denoted as zfemale, contains standardised values with a mean of zero and a standard deviation of one.

```
# Compute mean and standard deviation for each column
female_mean = np.mean(female, axis=0)
female_std = np.std(female, axis=0)

# Standardise the female dataset (z-scores)
zfemale = (female - female_mean) / female_std

# Verifying shape
zfemale.shape

zfemale_display = pd.DataFrame(zfemale, columns=column_names_bmi)
zfemale_display.head()
```

	Weight (kg)	Height (cm)	Upper arm length (cm)	Upper leg length (cm)	Arm circumference (cm)	Hip circumference (cm)	Waist circumference (cm)	BMI
0	0.914295	0.008950	-0.567395	1.132988	0.550845	1.083162	1.115785	0.996968
1	0.635776	-1.053068	-1.078931	-1.293492	1.032218	1.044755	0.265089	1.156175
2	-0.204423	0.150553	0.583563	0.261944	-0.162299	-0.190647	-0.372933	-0.259275
3	-0.728968	-0.387537	0.839332	-0.764643	-0.661500	-0.523501	-0.459152	-0.670391

The standardised dataset has a mean of approximately zero and a standard deviation of one for each column, confirming successful standardisation. This transformation ensures that all variables contribute equally to subsequent analyses, such as correlation and scatterplot comparisons.

8. Scatterplot Matrix and Correlation Analysis

To examine relationships between key anthropometric variables, a scatterplot matrix is constructed using the standardised female dataset. Pearson's and Spearman's correlation coefficients are computed to quantify the strength and nature of these relationships.

```
# Extracting required variables from zfemale
# Select required variables from the standardised female dataset
z_height = zfemale[:, 1]
z_weight = zfemale[:, 0]
z_waist = zfemale[:, 6]
z_hip = zfemale[:, 5]
z_bmi = zfemale[:, 7]

# Creating the scatterplot matrix
variables = [z_height, z_weight, z_waist, z_hip, z_bmi]
labels = ["Height", "Weight", "Waist", "Hip", "BMI"]

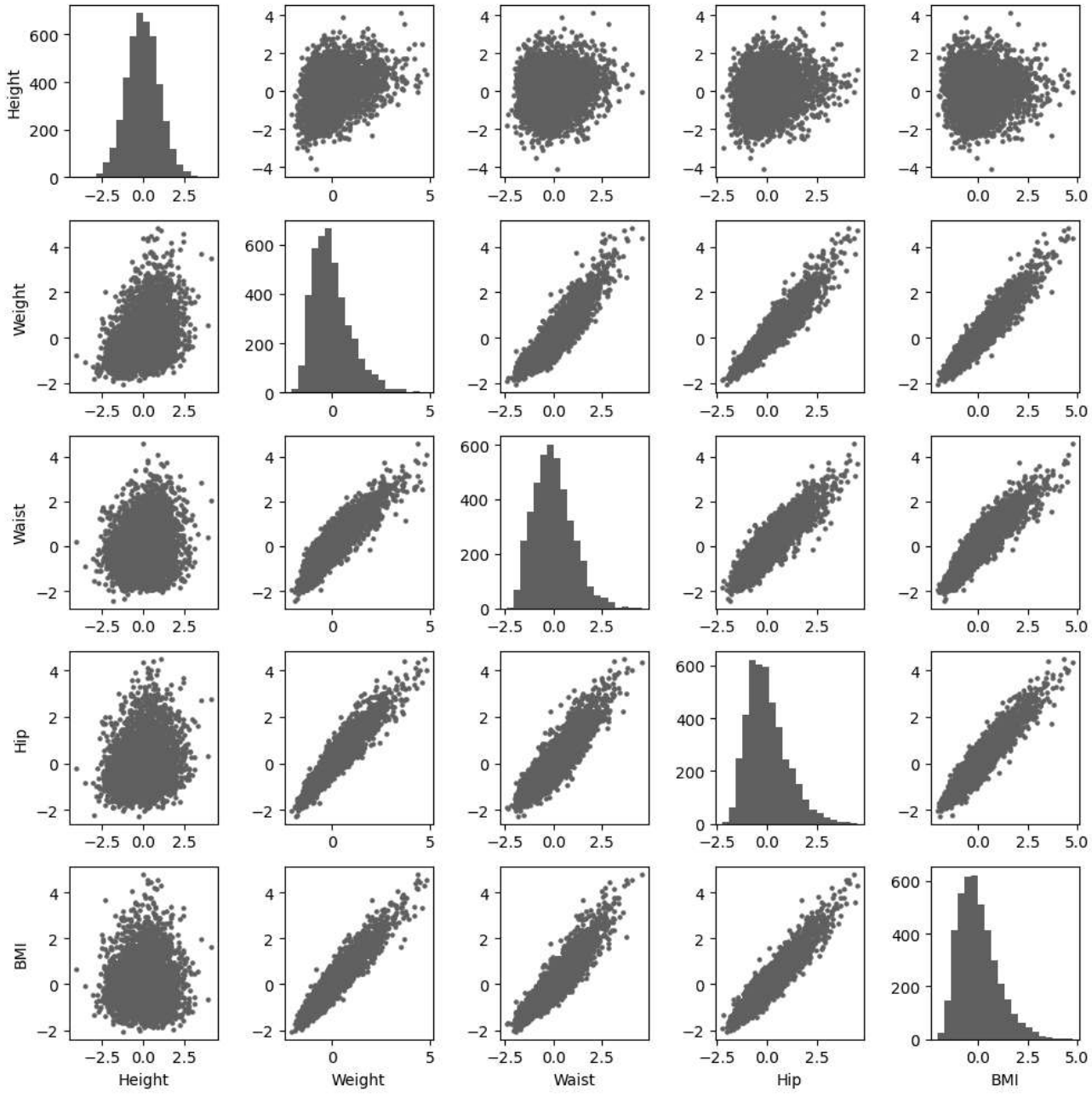
plt.figure(figsize=(10, 10))

for i in range(len(variables)):
    for j in range(len(variables)):
        plt.subplot(5, 5, i * 5 + j + 1)
        if i == j:
            plt.hist(variables[i], bins=20)
```

```
        else:
            plt.scatter(variables[j], variables[i], s=5)

        if i == 4:
            plt.xlabel(labels[j])
        if j == 0:
            plt.ylabel(labels[i])

plt.tight_layout()
plt.show()
```




```
from scipy.stats import pearsonr, spearmanr

data = np.column_stack([z_height, z_weight, z_waist, z_hip, z_bmi])
labels = ["Height", "Weight", "Waist", "Hip", "BMI"]

# Compute correlations
for i in range(len(labels)):
    for j in range(i + 1, len(labels)):
        pearson_corr, _ = pearsonr(data[:, i], data[:, j])
        spearman_corr, _ = spearmanr(data[:, i], data[:, j])
```

```
            print(f"{labels[i]} vs {labels[j]}:")
            print(f"  Pearson correlation = {pearson_corr:.2f}")
            print(f"  Spearman correlation = {spearman_corr:.2f}\n")
```

```
Height vs Weight:
  Pearson correlation = 0.35
  Spearman correlation = 0.34

Height vs Waist:
  Pearson correlation = 0.13
  Spearman correlation = 0.11

Height vs Hip:
  Pearson correlation = 0.20
  Spearman correlation = 0.21

Height vs BMI:
  Pearson correlation = 0.03
  Spearman correlation = 0.02

Weight vs Waist:
  Pearson correlation = 0.90
  Spearman correlation = 0.90

Weight vs Hip:
  Pearson correlation = 0.95
  Spearman correlation = 0.95

Weight vs BMI:
  Pearson correlation = 0.95
  Spearman correlation = 0.94

Waist vs Hip:
  Pearson correlation = 0.90
  Spearman correlation = 0.89

Waist vs BMI:
  Pearson correlation = 0.92
  Spearman correlation = 0.92

Hip vs BMI:
  Pearson correlation = 0.94
  Spearman correlation = 0.93
```

The scatterplot matrix reveals strong positive relationships between weight, waist circumference, hip circumference, and BMI. Pearson's and Spearman's correlation coefficients are largely consistent, indicating that these relationships are both linear and monotonic. Height exhibits comparatively weaker correlations with BMI, suggesting that body mass is more strongly influenced by weight and waist-related measurements.

9. Waist-Based Ratios

Two additional health indicators, the waist-to-height ratio and the waist-to-hip ratio, are computed for both male and female participants. These ratios provide insight into central fat distribution and are commonly used to assess obesity-related health risks.

```
# Re-initialize female array from original DataFrame to prevent multiple appends
female = female_df.to_numpy()

# Extract weight and height from the female matrix
female_weight = female[:, 0]
female_height_cm = female[:, 1]

# Convert height from cm → meters
female_height_m = female_height_cm / 100

# Computing BMI
female_bmi = female_weight / (female_height_m ** 2)

# Append BMI as the 8th column (re-adding in case this cell is run independently)
female = np.column_stack((female, female_bmi))

# Female ratios
female_waist = female[:, 6]
female_height = female[:, 1]
female_hip = female[:, 5]
```

```
female_waist_height_ratio = female_waist / female_height
female_waist_hip_ratio = female_waist / female_hip

# Append ratios to female matrix
female = np.column_stack((female, female_waist_height_ratio, female_waist_hip_ratio))

female.shape

female_column_names = [
    "Weight (kg)", "Height (cm)", "Upper arm length (cm)",
    "Upper leg length (cm)", "Arm circumference (cm)",
    "Hip circumference (cm)", "Waist circumference (cm)",
    "BMI", "Waist-to-Height Ratio", "Waist-to-Hip Ratio"
]

pd.DataFrame(female, columns=female_column_names).head()
```

	Weight (kg)	Height (cm)	Upper arm length (cm)	Upper leg length (cm)	Arm circumference (cm)	Hip circumference (cm)	Waist circumference (cm)	BMI	Waist-to-Height Ratio	Waist-to-Hip Ratio
0	97.1	160.2	34.7	40.8	35.8	126.1	117.9	37.835041	0.735955	0.934972
1	91.1	152.7	33.5	33.0	38.5	125.5	103.1	39.069720	0.675180	0.821514
2	73.0	161.2	37.4	38.0	31.8	106.2	92.0	28.092655	0.570720	0.866290
3	61.7	157.4	38.0	34.7	29.0	101.0	90.5	24.904378	0.574968	0.896040

```
# Re-initialize male array from original DataFrame to prevent multiple appends
male = male_df.to_numpy()

# Male ratios
male_waist = male[:, 6]
male_height = male[:, 1]
male_hip = male[:, 5]

male_waist_height_ratio = male_waist / male_height
male_waist_hip_ratio = male_waist / male_hip

# Append ratios to male matrix
male = np.column_stack((male, male_waist_height_ratio, male_waist_hip_ratio))

male.shape

male_column_names = [
    "Weight (kg)", "Height (cm)", "Upper arm length (cm)",
    "Upper leg length (cm)", "Arm circumference (cm)",
    "Hip circumference (cm)", "Waist circumference (cm)",
    "Waist-to-Height Ratio", "Waist-to-Hip Ratio"
]

pd.DataFrame(male, columns=male_column_names).head()
```

	Weight (kg)	Height (cm)	Upper arm length (cm)	Upper leg length (cm)	Arm circumference (cm)	Hip circumference (cm)	Waist circumference (cm)	Waist-to-Height Ratio	Waist-to-Hip Ratio
0	98.8	182.3	42.0	40.1	38.2	108.2	120.4	0.660450	1.112754
1	74.3	184.2	41.1	41.0	30.2	94.5	86.8	0.471227	0.918519
2	103.7	185.3	47.0	44.0	32.0	107.8	109.6	0.591473	1.016698
3	86.0	167.8	39.5	38.4	29.0	106.4	108.3	0.645411	1.017857

The waist-to-height and waist-to-hip ratios were successfully computed and appended to both datasets. These ratios provide additional insight into body fat distribution and enable more meaningful comparisons of central obesity between male and female participants.

10. Comparison of Waist-Based Ratios

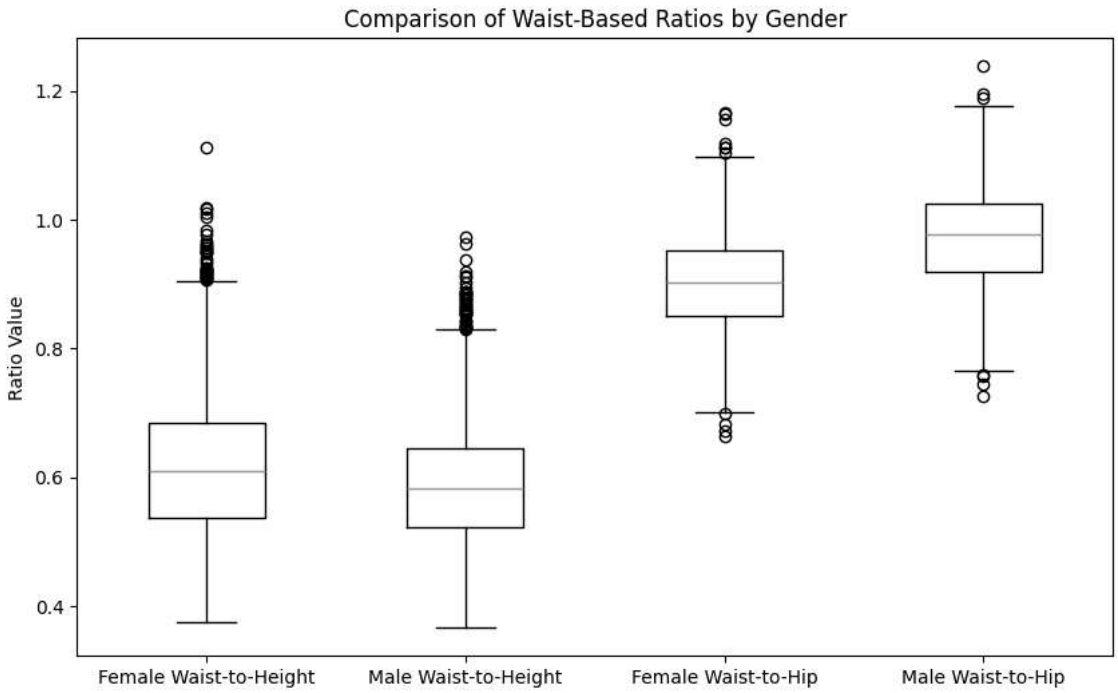
To compare central fat distribution between male and female participants, a box-and-whisker plot is constructed for the waist-to-height and waist-to-hip ratios. This visualization highlights differences in central tendency, variability, and outliers across gender and ratio type.

```
# Extract ratios
female_wht = female[:, 8]
female_whr = female[:, 9]

male_wht = male[:, 7]
male_whr = male[:, 8]

plt.figure(figsize=(10, 6))
plt.boxplot(
    [female_wht, male_wht, female_whr, male_whr],
    labels=[
        "Female Waist-to-Height",
        "Male Waist-to-Height",
        "Female Waist-to-Hip",
        "Male Waist-to-Hip"
    ]
)
plt.ylabel("Ratio Value")
plt.title("Comparison of Waist-Based Ratios by Gender")
plt.show()
```

```
/tmp/ipython-input-3251571181.py:9: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_lab
  plt.boxplot(
```



The boxplot reveals clear differences in waist-based ratios between male and female participants. Male participants generally exhibit higher waist-to-height and waist-to-hip ratios, indicating greater central fat accumulation. Waist-to-height ratios show less variability compared to waist-to-hip ratios, suggesting that they may provide a more stable indicator of central obesity.

11. Advantages and Limitations of Body Measurement Indicators

Body Mass Index (BMI)

Advantages:

- Simple to compute using weight and height.
- Widely used and easily comparable across populations.
- Useful for large-scale screening and epidemiological studies.

Limitations:

- Does not distinguish between fat mass and muscle mass.
- May misclassify muscular individuals as overweight.
- Does not capture fat distribution or central obesity.

Waist-to-Height Ratio

Advantages:

- Strongly associated with central obesity and cardiovascular risk.
- Accounts for individual height differences.
- Simple to interpret and calculate.

Limitations:

- Does not distinguish between subcutaneous and visceral fat.
- Measurement accuracy depends on correct waist measurement.
- May vary across age groups.

Waist-to-Hip Ratio

Advantages:

- Provides information on fat distribution patterns.
- Useful for assessing abdominal obesity.
- Can indicate health risks not captured by BMI.

Limitations:

- Sensitive to measurement errors in waist and hip circumference.
- Less effective in individuals with uniformly distributed fat.
- Interpretation may vary across populations.

12. Analysis of Extreme BMI Values

To examine body measurement patterns associated with extreme BMI values, the standardised measurements of the five individuals with the lowest BMI and the five individuals with the highest BMI are extracted and analyzed.

```
# Extracting BMI values (from female dataset)
female_bmi = female[:, 7]

# Sorting BMI indices
sorted_bmi_indices = np.argsort(female_bmi)

# Selecting lowest & highest BMI indices
lowest_bmi_indices = sorted_bmi_indices[:5]
highest_bmi_indices = sorted_bmi_indices[-5:]

# Extracting standardised rows from zfemale
lowest_bmi_z = zfemale[lowest_bmi_indices]
highest_bmi_z = zfemale[highest_bmi_indices]

# Display of results
zfemale_columns = [
    "Weight", "Height", "Upper arm length", "Upper leg length",
    "Arm circumference", "Hip circumference", "Waist circumference",
    "BMI"
]

print("Standardised measurements for lowest BMI individuals:")
pd.DataFrame(lowest_bmi_z[:, :8], columns=zfemale_columns)

print("\nStandardised measurements for highest BMI individuals:")
pd.DataFrame(highest_bmi_z[:, :8], columns=zfemale_columns)
```

```
Standardised measurements for lowest BMI individuals:

Standardised measurements for highest BMI individuals:
```

	Weight	Height	Upper arm length	Upper leg length	Arm circumference	Hip circumference	Waist circumference	BMI
0	4.247240	0.292155	1.862405	-0.982404	2.369364	4.104456	3.817318	4.396492
1	4.456130	0.504559	1.691893	-1.137948	3.349938	3.982836	2.903395	4.462011
2	4.349364	0.277995	2.842851	1.941815	4.366169	3.918826	3.754091	4.514669
3	4.804279	0.886886	2.118174	1.817380	3.777825	4.021242	4.081723	4.542709
4	4.363290	-0.033530	-0.055858	-0.173578	2.779422	4.315691	4.570299	4.762433

Individuals with the highest BMI values exhibit strongly positive standardised measurements for weight, waist circumference, and hip circumference, indicating substantially above-average body mass and central fat accumulation. Conversely, individuals with the lowest BMI values show negative or near-zero standardised measurements, reflecting below-average body mass across multiple anthropometric variables.

Conclusion

In this report, body measurement data from the NHANES survey were analyzed to compare anthropometric characteristics between adult male and female participants. Using NumPy-based numerical analysis and Matplotlib visualizations, differences in body weight distributions, variability, and central tendency were identified and systematically examined.

Male participants generally exhibited higher body weights and greater variability, as supported by both visual (histograms and boxplots) and numerical analyses. Standardisation of the female dataset enabled meaningful comparison between variables measured on different scales and facilitated correlation analysis. Strong positive relationships were observed between weight, waist circumference, hip circumference, and BMI, highlighting the importance of central fat accumulation in overall body mass.

Waist-to-height and waist-to-hip ratios provided additional insight into fat distribution patterns and revealed gender-based differences in central obesity. Compared to BMI, waist-based ratios offered a more direct measure of abdominal fat, though each metric has distinct advantages and limitations.

Finally, the analysis of individuals with extreme BMI values demonstrated clear patterns in standardised body measurements, reinforcing the relationship between BMI and waist-related measurements. Overall, this study illustrates how numerical summaries, visual exploration, and derived health indicators can be combined to gain a comprehensive understanding of body composition data.